

Friendly reminder:

Homework problems are meant to help study the lecture contents and will not count towards final grade. Points are given for feedback only.

The final grade will be formed from the mid-term exam (Nov 4, 40%) and the final exam (Dec 30, 60%).

2nd Homework sheet (due Sept 30)

k always denotes an algebraically closed field of characteristic 0.

Problem 1 (Topological properties, 3P)

Let $f : X \rightarrow Y$ be a continuous map of topological spaces. Show that if X has one of the properties quasi-compact/connected/irreducible, then $f(X)$ has the same property.

Problem 2 (Irreducible components, 3P + 2P)

Let X be a quasi-compact topological space that can be covered by open subsets homeomorphic to $\mathrm{Sp}(A)$ for varying reduced finite type k -algebras A .

- (a) Show that any descending chain $X \supseteq Z_1 \supseteq Z_2 \supseteq \dots$ of closed subsets of X stabilizes.
- (b) Deduce that every closed subset $Z \subseteq X$ can be written as a finite union of irreducible closed subsets.

Problem 3 (Centralizers, 3P + 2P + 3P)

- (a) Let S , X and Y be affine k -varieties, and let $f : X \rightarrow S$ and $g : Y \rightarrow S$ be morphisms. Show that

$$X \times_S Y := \{(x, y) \in X \times Y \mid f(x) = g(y)\}$$

is a closed subset of $X \times Y$. In particular, it is again naturally an affine variety.

- (b) Let G be a LAG over k and let $x \in G$ be an element. Use (a) to show that the centralizer

$$G_x := \{g \in G \mid gx = xg\}$$

is a closed subgroup of G .

- (c) Determine the centralizers of

$$\begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \in \mathrm{SL}_2 \quad \text{and} \quad \begin{pmatrix} 2 & & \\ & 2 & \\ & & 1 \end{pmatrix} \in \mathrm{GL}_3.$$

Problem 4 (Centers, 2P + 3P)

- (a) Building on the previous exercise, show that the center of an algebraic group is a closed subgroup.
- (b) Determine the center of the subgroup of upper triangular matrices $T_n \subseteq \mathrm{GL}_n$. Express it as a product of copies of $\mathbb{G}_m$ and $\mathbb{G}_a$.